PKU–Zurich PhD Summer School on Machine Learning for Macroeconomics and Finance

Optimal carbon taxes with deep surrogates — solve once, ask many questions

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Roadmap of This Lecture

1. **Where Session 2 left us.** One trained surrogate, one question answered (estimation); two questions still open.
2. **The two-surrogate pipeline, generalized.** Pseudo-states $(\boldsymbol{\theta}, \vartheta)$, quantities of interest that generalize $m(\theta)$, and three post-processing operators.
3. **Design: constrained optimal carbon taxes.** Taxes and transfers in a stochastic OLG IAM: three policy scenarios, from unconstrained to Pareto-improving.
4. **Synthesis.** One surrogate, three questions; practical guidance; extensions and open questions.
5. **Appendix: Quantify.** Uncertainty quantification of the social cost of carbon (univariate effects, Sobol' indices, Shapley values), plus model and calibration details.

Learning objectives

By the end of this lecture you should be able to:

- Extend pseudo-state surrogates from estimation to uncertainty quantification and policy design
- Compute the social cost of carbon and welfare quantities of interest from surrogate simulations
- Rank parameter uncertainties via univariate effects, Sobol' indices, and Shapley values
- Add a Gaussian process and Bayesian-active-learning layer when each QoI evaluation needs heavy Monte Carlo
- Solve constrained policy problems, including Pareto constraints, directly on the surrogate

Where Session 2 Left Us

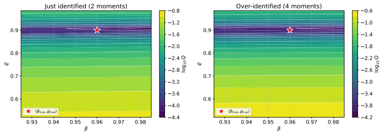
This morning's takeaways:

- A neural surrogate with parameters as *pseudo-states* decouples the cost of solving the structural model from the cost of moment-matching.
- Train once, use it inside any optimizer: millions of cheap moment evaluations under common random numbers.
- Joint estimation made the *identification ridge* visible; the interior-solution check closed the loop.

The pivot: estimation treats $\boldsymbol{\theta}$ as unknown but **fixed**. Two questions remain:

- What if $\boldsymbol{\theta}$ **stays uncertain** after all we know?
- What if some inputs are not parameters at all, but **choices**?

Today's mantra: **solve once, ask many questions.**



Session 2's identification ridge: the criterion's shape, not its argmin, carried the lesson.

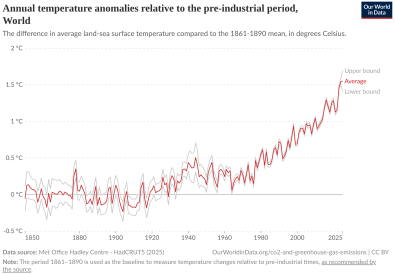
Three Operators, One Surrogate

Three questions ride on one parametric surrogate. Session 2 asked the first; today we work out **design** in full — the appendix works out **quantify**.

Operator	Question	Post-processing	Where
Estimate	Which parameters fit the data?	$\arg\min$ of an SMM criterion over $\boldsymbol{\theta}$	Session 2
Quantify	Which parameter uncertainties matter?	propagate $p(\boldsymbol{\theta})$ through a QoI: univariate effects, Sobol' indices, Shapley values	Appendix (SCC)
Design	Which policy is best?	constrained $\arg\max$ of a welfare QoI over ϑ	Today (carbon taxes)

Identical machinery throughout: the **two-surrogate pipeline**. Surrogate 1 solves the equilibrium over states and pseudo-states; Surrogate 2 maps parameters to quantities of interest. Only the post-processing operator changes.

The Challenge: Climate Risk & Heterogeneity





- **Systemic Risk:** Climate change poses a fundamental threat to global economic stability and human welfare.
- **Non-Linear Dynamics:** Tipping points introduce the risk of irreversible, catastrophic environmental shifts.
- **Distributional Impact:** Effects will be highly heterogeneous, creating profound equity challenges across regions and generations.
- **Literature Context:** These macro-financial linkages are detailed in the recent review Fernandez-Villaverde, Gillingham, and Scheidegger (2025).

Two Policy Questions, One Computational Wall

Quantify: how uncertain is the SCC? (worked out in the appendix)

- **Carbon taxation**: widely supported policy response to the climate change problem.
- The level of the carbon tax is based on the social cost of carbon (SCC): the marginal loss caused by an extra ton of CO_2 emissions.
- SCC values are computed with **integrated assessment models** (IAMs) that link economy and climate.
- IAMs are subject to **significant parametric uncertainty, model uncertainty, and climate risks of tipping points**.

Design: which carbon tax is best?

- **The Consensus:** Carbon pricing is widely regarded as the most efficient tool to mitigate emissions.
- **The Difficulty:** In a world with heterogeneous agents (e.g., OLG), simple taxes create winners and losers, threatening political feasibility.
- **The Objective:** A systematic way to compute constrained optimal taxes that account for stochastic risks (tipping points) and respect political constraints (Pareto improvements): most work relies on simplifying assumptions or heuristic "guessing".

The wall: both questions need thousands of model evaluations. IAMs are complex non-linear models highly susceptible to the curse of dimensionality; evaluating expected welfare requires 10,000 Monte Carlo paths per policy ϑ. Session 2's answer carries over: with a deep surrogate, the inner loop is three orders of magnitude cheaper.

The Two-Surrogate Pipeline in a Nutshell

- **Step 1: Global solution with DEQN (Surrogate 1)**
  - Solve the model as a function of states and pseudo-states in one go: train on $\tilde{x} = (x, \boldsymbol{\theta}, \vartheta)$, exactly the pseudo-state trick of Session 2.
  - For linear tax rules in cumulative emissions, κ, *TP*: 4 tax pseudo-states $\vartheta = (\vartheta_0, \vartheta_E, \vartheta_\kappa, \vartheta_{TP})$, plus 12 from transfer shares.

  $\Rightarrow$ Once trained, the solution allows us to simulate data essentially for free.
- **Step 2: Simulate quantities of interest (Surrogate 2 where needed)**
  - $\widehat{\mathrm{QoI}}(\cdot)$ generalizes Session 2's moment map $m(\theta)$: in Session 2 three moments; today 40 cohort welfares, and in the appendix the SCC.
  - Where one QoI evaluation needs heavy Monte Carlo, fit a Gaussian process with Bayesian active learning over the parameters.[1]
- **Step 3: Post-process**
  - Estimate (arg min), quantify (global sensitivity analysis), or design (constrained arg max): three operators on the same object.

[1] A surrogate is a 21st-century lookup table: $\mathrm{Welfare}(\vartheta_1, \cdots, \vartheta_N)$.

Recall: Steps of the Method (2 Surrogates)

Step 1: DEQN — $P(x_{\text{state},t}, \vartheta)$

$\downarrow$

Step 2: $\text{GP}(\theta)$ — $\widehat{\text{QoI}}(\vartheta)$

$\downarrow$

Step 3: Optimizer — $\arg\max \widehat{\text{QoI}}(\vartheta)$

The final box branches into the three operators: **estimate** (arg min, Session 2) · **quantify** (global sensitivity analysis, appendix) · **design** (constrained arg max, today).

Recap: Deep Equilibrium Nets (assumed known)

A **functional rational expectations equilibrium**: $\{f_i\}_{i=1}^{N_{\text{out}}}$, where

$$f_i : \mathcal{D} \subset \mathbb{R}^{N_{\text{in}}} \to \mathbb{R} : \underbrace{x}_{\text{state}} \to \underbrace{f_i(x)}_{\text{endogenous variables}}, \text{ s.t. : } \underbrace{\mathbf{G}(x, f_1, \ldots, f_{N_{\text{out}}}) = 0}_{\text{equilibrium conditions}}$$

A **deep equilibrium net**: $\mathcal{N}_\rho$, with $\mathcal{N}_\rho : \mathcal{D} \subset \mathbb{R}^{N_{\text{in}}} \to \mathbb{R}^{N_{\text{out}}}$, trained so that $\mathcal{N}_\rho(x) \approx (f_1(x), \ldots, f_{N_{\text{out}}}(x))$ by minimizing the implied error in the equilibrium conditions along simulated paths (Azinovic et al., 2022).[2]

Callback to Session 2

In Session 2 the QoI was the moment vector $m(\theta)$ and the operator was $\arg\min$; today the QoIs are the SCC and cohort welfares, and the operators change.

Notation. On Day 2, θ denoted the network weights; today $\boldsymbol{\theta}$ is reserved for structural parameters, so the network weights are ρ.

[2]Sample codes: `https://github.com/sischei/DeepEquilibriumNets`. Loss function, training loop, and architecture details: appendix.

What Scales Up Relative to Session 2

	Session 2 (Brock–Mirman)	Today (climate applications)
Surrogate 1 inputs	(z, K, ϱ), then (z, K, β, ϱ)	OLG IAM state $[t, TP_t, TP_{\text{reached}}, \kappa_t, a_{t,j}, E_t]$ plus 16 policy pseudo-states; IAM state plus uncertain $\boldsymbol{\theta}$ in the UQ appendix
QoI shape	three scalar moments m_1, m_2, m_3	an SCC surface $\mathrm{SCC}(\theta_1, \ldots, \theta_N)$ and 40 cohort welfares
Second surrogate	none needed: moments were cheap	GP regression + Bayesian active learning: each welfare evaluation costs 10,000 simulated paths

Cost that does not scale up: Surrogate 1 trains in about 4h from scratch on a laptop, minutes when warm-started; Surrogate 2 needs only 500 simulations (450 Latin hypercube + 50 actively learned).

Post-Processing: Three Operators, Formally

Estimate (Session 2, for continuity).

$$\widehat{\theta}_{\text{SMM}} = \arg\min_{\theta} \left[\widehat{g}_T - \widehat{m}_S(\theta)\right]^\top W \left[\widehat{g}_T - \widehat{m}_S(\theta)\right]$$

Quantify (appendix). For a QoI $Y = \mathcal{M}(\boldsymbol{\theta})$ with uncertain $\boldsymbol{\theta} \sim p(\boldsymbol{\theta})$:

$$\underbrace{\mathcal{M}_i^1(t) = \mathbb{E}_{\theta_{-i}}[Y \mid \theta_i = t]}_{\text{univariate effect}} \qquad \underbrace{S_i = \frac{\mathrm{Var}_{\theta_i}[\mathbb{E}\,[Y \mid \theta_i]]}{\mathrm{Var}\,[y]}, \quad S_i^T = \frac{\mathbb{E}_{\theta_{-i}}[\mathrm{Var}\,[Y \mid \theta_{-i}]]}{\mathrm{Var}\,[y]}}_{\text{first-order and total Sobol' indices}} \qquad \underbrace{Sh_i}_{\substack{\text{Shapley value: fair attribution} \\ \text{of } \mathrm{Var}(\mathrm{QoI}) \text{ across parameters}}}$$

Design (today). For a welfare QoI over policy parameters ϑ:

$$\vartheta^* = \arg\max_{\vartheta} \widehat{\mathrm{QoI}}(\vartheta) \quad \text{s.t. feasibility and, if imposed, Pareto constraints.}$$

Same surrogate under all three operators; only the question changes.

Today's Instantiations

Main part: Design

ϑ = tax and transfer parameters · QoI = 40 cohort welfares · operator = constrained $\arg\max$.

Appendix: Quantify

$\boldsymbol{\theta}$ = structural parameters (preferences, damages, climate sensitivity) · QoI = the social cost of carbon · operator = global sensitivity analysis.

Optimal Carbon Taxes and Transfers

Overview of our Stochastic IAM

- 12 Overlapping generations of selfish agents.
- 5-yearly-calibrated model: agents live from age 20 to age 80.
- Use a simple climate module (Dietz et al., 2021).
- Use a damage function with stochastic tipping (Kotlikoff et al., 2021b).
- Exogenously declining carbon intensity as in DICE-2016 (Nordhaus, 2017a).

Technology

Firms, in each period, t, produce final output, Y_t, and emissions e_t with capital, K_t and labor, L_t, and choose how much of its emissions to mitigate μ_t according to

$$(Y_t, e_t) = (\Omega_t(T_t)\Phi(\mu_t)K_t^{\alpha}L_t^{1-\alpha} + (1-\delta)K_t,\ \kappa_t(1-\mu_t)K_t^{\alpha}L_t^{1-\alpha}) \tag{1}$$

- $\Omega_t(T_t)$: climate damages that reduce output.
- $\Phi(\mu_t)$ is a mitigation cost function.
- α is the capital share.
- κ_t is carbon intensity. We assume that κ_t decreases stochastically over time as in Kotlikoff et al. (2021b).
- The abatement technology (adapted from Nordhaus, 2017b): $\Phi(\mu_t) = (1-\theta_1\mu_t^{\theta_2})$.
- θ_1 and θ_2 are parameters and $\mu_t \in [0,1]$.

Profit maximization is given by:

$$\begin{aligned} \max_{\mu_t, K_t, L_t} \quad & \Omega_t(T_t)(1-\theta_1\mu_t^{\theta_2})K_t^{\alpha}L_t^{1-\alpha} + (1-\delta)K_t - w_tL_t - (1+r_t)K_t - \tau_t(1-\mu_t)\kappa_t K_t^{\alpha}L_t^{1-\alpha} \\ \text{s.t.} \quad & 0 \le \mu_t \le 1. \end{aligned} \tag{2}$$

Households

- Agents enter the labor force at age 20 with 0 assets and die at age 80, after 12 periods.
- Agents born in period t maximize lifetime utility:

$$\mathbb{E}_t \sum_{j=1}^{12} \beta^j \frac{C_{t+j-1,j}^{1-\sigma}}{1-\sigma}, \tag{3}$$

$$\text{subject to } C_{t,j} + a_{t+1,j+1} = (1+r_t)a_{t,j} + w_t l_j + \mathbb{T}_{t,j}, \tag{4}$$

  where $C_{t,j}$, $a_{t,j}$, l_j and $\mathbb{T}_{t,j}$ reference, respectively, consumption, asset level, labor endowment, and transfer of those born at time t who are currently age j.
- l_j is similarly calibrated as in Kotlikoff et al. (2021a).
- Agents work for 8 periods, earning labor income, and then retire. Agents aged 9 to 12 work on a 40% basis as a proxy for a state pension system.
- β is the discount factor and σ is the coefficient of relative risk aversion.
- Welfare criterion is

$$U_t = \mathbb{E}_0 \sum_{j=1}^{12} \beta^j \frac{C_{t+j-1,j}^{1-\sigma}}{1-\sigma}$$

  Use $\tilde{U}_t$ for welfare under taxation.

Climate Block: TCRE and Tipping Damages

Temperature: near-linear in cumulative emissions.

- Global mean temperature increase (T_t^{AT}) is proportional to cumulative CO_2 emissions (E_s):

$$T_t^{AT} \approx \sigma_{CCR} \sum_{s=0}^{t} E_s \tag{5}$$

- The proportionality constant σ_{CCR}, the Carbon-Climate Response (CCR) or TCRE, ranges from 1.0 °C to 2.3 °C per 1 000 GtC.
- Efficient and widely used in IAMs; by itself it does not account for uncertainties in σ_{CCR} nor for nonlinear effects such as tipping points (cf. Dietz and Venmans, 2019).

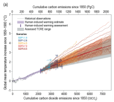


Figure from IPCC 6.

Damages: stochastic climate tipping.

- Damages reduce output; stochastic tipping as in Weitzman (2012); Kotlikoff et al. (2021b): irreversible damage to the environment (e.g., Lenton et al., 2008).
- Damage function:

$$\Omega_t(T_t) = \frac{1}{1 + \left(\frac{1}{\psi_1} T_t\right)^2 + \left(\frac{1}{2 \cdot TP_t} T_t\right)^{6.754}} \tag{6}$$

where ψ_1 is a damage parameter and TP_t follows a random walk truncated at $TP_t \in [2.5, 3.5]$.

- Once the temperature hits a tipping point, TP_t stays at the given level.
- Shock processes for κ_t and TP_t: appendix.

Equilibrium Conditions

This system defines the DEQN loss for today's application.

Firm:

$$0 = \tau_t \kappa_t - \Omega_t(T_{\mathrm{AT},t})\theta_1\theta_2\mu_t^{\theta_2 - 1} \qquad (\mu_t)$$

$$r_t = \left(\Omega_t(T_{\mathrm{AT},t})(1-\theta_1\mu_t^{\theta_2}) - \tau_t\kappa_t(1-\mu_t)\right)\alpha K_t^{\alpha-1}L_t^{1-\alpha} - \delta \qquad (K_t)$$

$$w_t = \left(\Omega_t(T_{\mathrm{AT},t})(1-\theta_1\mu_t^{\theta_2}) - \tau_t\kappa_t(1-\mu_t)\right)(1-\alpha)K_t^{\alpha}L_t^{-\alpha} \qquad (L_t)$$

Households:
For all households $j \in 1, ..., A-1$:

$$c_t^{-\sigma} = \beta\mathbb{E}_t\left[(1+r_{t+1})c_{t+1,j+1}^{-\sigma}\right] \qquad (\mathrm{EE})$$

$$C_{t,j} + a_{t+1,j+1} = (1+r_t)a_{t,j} + w_t l_{t,j} + \mathbb{T}_{t,j} \qquad (\mathrm{BC})$$

and initial/final conditions $a_{t,1} = 0$, $a_{t+1,A} = 0$.

Market Clearing:

$$K_t = \sum_{j=1}^{A} a_{t,j} \qquad (K_t)$$

$$L_t = \sum_{j=1}^{A} l_j \qquad (L_t)$$

$$\tau_t e_t = \sum_{j=1}^{A} \mathbb{T}_{t,j} \qquad (\mathbb{T})$$

The Planner's Problem

- **Objective:** The planner chooses policy parameters ϑ (taxes & transfers) to maximize the Social Welfare Function $\mathcal{W}$:

$$\max_{\vartheta} \quad \mathcal{W}(\vartheta) = \sum_{t=-10}^{30} \gamma_t \tilde{U}_t(\vartheta) \tag{7}$$

$$\text{s.t.} \quad \tilde{U}_t(\vartheta) \geq U_t \quad \forall t \quad \text{(if Pareto constrained)}$$

(Where $\tilde{U}_t(\vartheta)$ is the ex-ante expected lifetime utility of generation t)

- **Welfare Metric:** Gains are reported as the consumption equivalence factor (CE_t):

$$\text{CE}_t = \left(\frac{\tilde{U}_t}{U_t}\right)^{\frac{1}{1-\sigma}} - 1 \quad \text{(where } U_t \text{ is BAU utility)}$$

- **Scope & Challenge:**
  - Horizon: 40 generations (11 current, 29 future).
  - Complexity: Solving for the optimal high-dimensional vector ϑ in a stochastic setting.

Mapping the OLG Model to DEQN

Goal: Approximate the policy functions of the OLG model using a neural network: the $(z, K, \varrho) \rightarrow s$ network of Session 2, scaled up.

Inputs (state vector):

$$x = \left[t,\ TP_t,\ TP_{\text{reached}},\ \kappa_t,\ a_{t,j},\ E_t,\ \vartheta\right] \in \mathbb{R}^{A+5+d_\vartheta}$$

Outputs (policy vector):

$$y = \left[a_{t+1,j+1},\ v_{t,j}\right] \in \mathbb{R}^{2(A-1)}$$

Policy map:

$$\Pi : x \ \Rightarrow\ y$$

Pseudo-states ϑ:

- $d_\vartheta = 16$: 4 tax parameters + 12 transfer shares.
- DEQN learns policies for all tax rules and transfers at once.

Performance:

- Runtime from scratch: ≈4h on laptop.
- Warm start: a few minutes.
- Accuracy: MSE $\sim 10^{-6}$, EE $\sim 10^{-3}$.

Approximating Welfare with Surrogates

The Computational Bottleneck:

- DEQN solves the equilibrium, but evaluating expected welfare $\mathbb{E}[\mathcal{W}]$ requires expensive Monte Carlo simulations (10,000 paths per policy ϑ).
- **Solution:** Train a Gaussian Process (GP) to map policy parameters ϑ directly to welfare outcomes.

Two Optimization Targets (QoIs):

1. Maximize Aggregate Welfare

Goal: Maximize scalar $\mathcal{W}$.

$$\max_{\vartheta} \mathbb{E}_0[\mathcal{W}(\vartheta)]$$

- **Surrogate:** Single scalar GP.
- **Implication:** Ignores distribution (allows losers).

2. Pareto Improvement

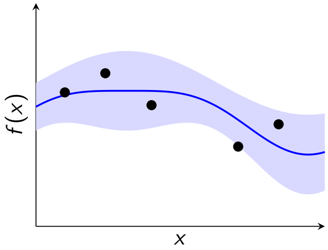
Goal: Maximize $\mathcal{W}$ subject to:

$$\tilde{U}_t(\vartheta) \geq U_t \quad \forall t$$

- **Surrogate:** Vector of 40 GPs (one per cohort).
- **Implication:** No generation is worse off.

The Engine: Gaussian Processes & Active Learning

1. Gaussian Processes (GP)



Kernel:

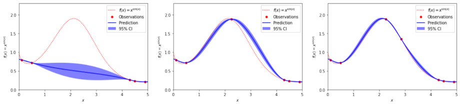
$$k(x, x') = \sigma_f^2 \exp\left(-\frac{\|x - x'\|^2}{2\ell^2}\right)$$

Predictive mean & variance:

$$\mu(x^*) = k_*^\top (K + \sigma_n^2 I)^{-1} \mathbf{y}$$
$$\sigma^2(x^*) = k(x^*, x^*) - k_*^\top (K + \sigma_n^2 I)^{-1} k_*$$

2. Bayesian Active Learning (BAL)

$$\alpha(x) = \underbrace{\mu(x)}_{\text{Exploit}} + \kappa \cdot \underbrace{\sigma(x)}_{\text{Explore}}$$



We simulate points that maximize this score, reducing computational cost by orders of magnitude.

Constructing the Welfare Surrogate

1. The Mapping

We map policy rules directly to **Expected Welfare**:

$$\vartheta \longmapsto \mathbb{E}_0[\mathcal{W}(\vartheta)]$$

- **Input (ϑ):** Tax and transfer parameters.
- **Target:** Monte Carlo mean of lifetime utilities.

2. Active Training

We populate the training set $\mathcal{D}$ efficiently:

- **Initial Design:** 450 points sampled randomly (Latin Hypercube).
- **Refinement:** 50 points added via Bayesian Active Learning (BAL).
- **Total:** Only 500 simulations needed.

3. The Model

Fitting the Gaussian Process:

- **Kernel:** Squared Exponential with Automatic Relevance Detection (ARD).
- **Accuracy:** Leave-One-Out (LOO) error $\approx \mathbf{10^{-4}}$.
- **Result:** A smooth, cheap-to-query surface.

Robustness. Training is anchored on a handful of trusted reference solutions to avoid *lazy learning* — the flat-in-the-parameter failure mode discussed in Session 1; 10–20 anchors suffice (Friedl et al., 2023).

Maximizing Welfare

- We now have a cheap-to-evaluate surrogate model to map tax functions onto welfare. But we still need to find the optimal tax.
- We aim to find the arg max of our surrogate model, which gives us the optimal tax function parameters:

$$\vartheta^* = \arg\max_{\vartheta} \widehat{\mathrm{QoI}}(\vartheta)$$

  s.t. various constraints.
- We use constrained optimization with linear constraints for periods $\{0,\mathrm{T}\}$ that ensure that the parameters of the tax function stay within the feasible range during optimization.

Callback to Session 2

Same object as the SMM criterion, opposite sign of intent: there we minimized distance to data, here we maximize welfare.

Three Policy Scenarios

- **Scenario 1: Unconstrained Welfare Maximization**
  Linear tax on cumulative emissions + exogenous transfers.

$$\tau_t(E_t) = \vartheta_0 + \vartheta_E \cdot E_t$$

- **Scenario 2: Pareto-Improving Policy**
  Linear tax on cumulative emissions + **optimal** transfers.

$$\tau_t(E_t) = \vartheta_0 + \vartheta_E \cdot E_t$$

- **Scenario 3: Increasing Policy Complexity**
  Linear tax on cumulative emissions, carbon intensity (κ), and tipping risk (TP) + optimal transfers.

$$\tau_t(\cdot) = \vartheta_0 + \vartheta_E E_t + \vartheta_\kappa \kappa_t + \vartheta_{TP}(1 - \mathbb{D}_{TP})$$

Scenario 1: Unconstrained Welfare Maximization

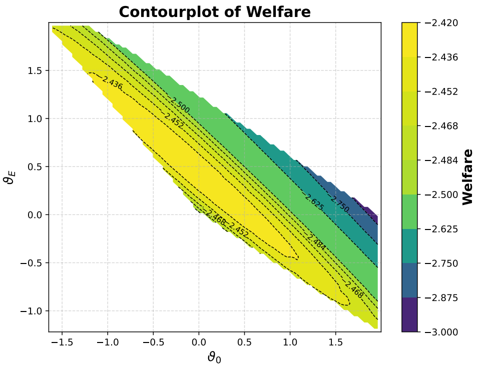
Each period's tax income is redistributed to currently alive agents; in general, transfer schemes may create winners and losers (Kotlikoff et al., 2021a). Here: the "10% scheme". The youngest generation receives the largest share of total tax revenue, and the share for each older generation decreases by 10% relative to the next younger one. The shares are normalized to sum to one. Formally, our transfer scheme, $\mathbb{T}$, is defined as:

$$\mathbb{T}_{t,j} = 0.9^{(j-1)} \cdot 0.1394 \cdot \tau_t e_t, \quad j \in \{2, ..., 12\}. \tag{8}$$

The resulting constrained optimization problem reads as:

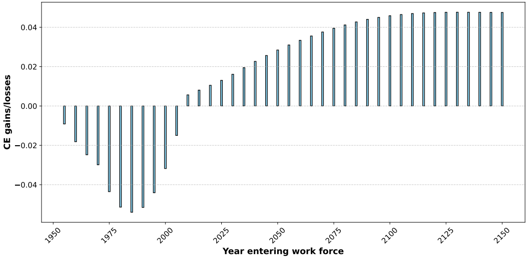
$$\begin{aligned} \vartheta^* &= \arg \max_{\vartheta \in [a,b]} \widehat{\mathrm{QoI}}(\vartheta) \\ &\quad \text{s.t.} \\ &\quad 0 \leq \vartheta_0 + \vartheta_E E_0 \leq \bar{\tau}, \\ &\quad 0 \leq \vartheta_0 + \vartheta_E E_{29} \leq \bar{\tau}. \end{aligned} \tag{9}$$

The Welfare Surrogate (Scenario 1)



- **The Surface:** A Gaussian Process approximation of the Social Welfare Function.
- **The Axes:** Tax Intercept (ϑ_0) vs. Slope (ϑ_E).
- **The Goal:** Find the “peak” (yellow zone) without running millions of simulations.
- **Precision:** Leave-One-Out error $\approx \mathbf{10^{-5}}$.

Optimal Welfare Distribution



The Price:

- **Aggregate Welfare Gain:** The total pie grows by **1.6%** (consumption equivalent).
- **Future Gain:** Generations born after 2100 enjoy gains of $\approx$ **4.8**%.
- **Current Pain:** This comes at a steep cost. Transition generations suffer losses up to **5%**.
- **Conclusion:** Without transfers, this optimal policy is politically dead on arrival.

Scenario 2: Pareto-Improving Policy

Lead-in: the welfare-maximizing tax reduces warming to 2.7°C, **but** imposes welfare losses of up to **5%** on current older generations: economically efficient, politically infeasible. **Goal:** a tax *and* transfer scheme where $\tilde{U}_t \geq U_t$ for all t.

For 40 values of t:

$$\mu_{*,t}(\vartheta) = \widehat{\mathrm{QoI}}_t(\vartheta) = \mathbb{E}\left[\tilde{U}_t(\vartheta)\right] \quad \forall\, t \in \{-10, \ldots, 29\} \tag{10}$$

Solve:

$$\begin{aligned} \vartheta^* &= \arg\max_{\vartheta \in [a,b]} \sum_{t=-10}^{29} \gamma_t \, \widehat{\mathrm{QoI}}_t(\vartheta) = \arg\max_{\vartheta \in [a,b]} \sum_{t=-10}^{29} \gamma_t \, \mathbb{E}_0\left[\tilde{U}_t(\vartheta)\right] \\ \text{s.t.} \quad & \mathbb{E}_0\left[\tilde{U}_t(\vartheta)\right] \geq \mathbb{E}_0\left[U_t\right] \;\; \forall\, t, \\ & 0 \leq \vartheta_0 + \vartheta_E E_0 \leq \bar{\tau}, \qquad 0 \leq \vartheta_0 + \vartheta_E E_{29} \leq \bar{\tau}, \qquad \sum_{i=1}^{12} \vartheta_i = 1 \end{aligned} \tag{11}$$

Optimal Parameters and Transfers

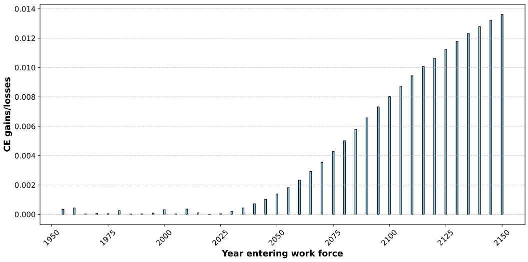
We find optimal tax parameters: $\vartheta_0 = -0.186$ and $\vartheta_E = 0.225$.

The optimal transfer shares allocate resources to specific cohorts to satisfy the Pareto constraints:

t1	**t2**	**t3**	**t4**	**t5**	**t6**	**t7**	**t8**	**t9**	**t10**	**t11**	**t12**
0.128	0.051	0.058	0.09	0.15	0.09	0.066	0.143	0.076	0.048	0.039	0.06

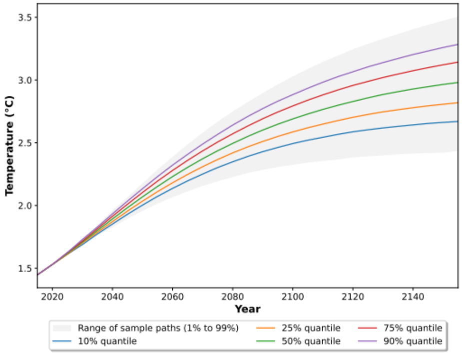
Table: Pareto optimal transfer scheme. Optimal shares for each concurrent cohort i under the linear cumulative emissions tax.

Optimal Welfare (Pareto Improving)



- **The Pareto Constraint:** Transfers are explicitly optimized to keep initial generations (and those born in the next decade) at **BAU levels** (0% loss).
- **Future Gains:** Welfare improves monotonically, reaching $\approx$**1.4%** for distant generations.
- **The Cost of Consensus:** Compared to the unconstrained case (max 4.8%), future gains are smaller, but **no generation loses**.
- **Aggregate:** Total welfare gain of **0.42%**.

Temperature (Scenario 2: Pareto-Improving)



- **The Trade-off:** Mean warming settles just below **3 °C** (higher than Scenario 1).
- **Crucial Insurance:** The 99th percentile **never exceeds 3.5 °C**.
- **Tail Truncation:** The policy deliberately eliminates the catastrophic tail ($>$3.6 °C under BAU).
- **The Win:** Achieved with optimal transfers $\rightarrow$ **no generation loses**.
- **Conclusion:** The Pareto-optimal policy functions primarily as disaster insurance rather than aggressive mitigation.

(Emissions distribution: appendix.)

Scenario 3: Increasing Policy Complexity

Question: Can we improve welfare by targeting specific climate drivers?

We expand the tax rule to include **Carbon Intensity** (κ_t) and **Tipping Risk** (TP_t):

$$\tau_t = \vartheta_0 + \vartheta_E \frac{E_t}{E_0} + \underbrace{\vartheta_\kappa \frac{\kappa_t}{\kappa_0}}_{\text{Intensity}} + \underbrace{\vartheta_{TP}(1 - \mathbb{D}_{TP})}_{\text{Tipping Risk}} \tag{12}$$

- $\mathbb{D}_{TP} \in [0, 1]$: Normalized distance to the tipping threshold.
- **Dimensionality:** This adds new pseudo-states, increasing the parameter space to $d_\vartheta = 16$ (4 tax parameters + 12 transfer shares).
- We use the same ML-based optimization to find the constrained Pareto-optimal policy.

Results: Diminishing Returns to Complexity

We find the optimal parameters:

$$\{\vartheta_0, \vartheta_E, \vartheta_\kappa, \vartheta_{TP}\} = \{-0.237,\ 0.203,\ 0.037,\ 0.012\}$$

Key Finding: Complexity adds very little welfare value.

Policy Specification	Instruments	Welfare Gain*
Linear in Cumulative Emissions (E_t)	2	**0.42%**
Linear in E_t, κ_t, TP_t	4	**0.45%**

Table: Aggregate welfare gains in consumption-equivalent terms.

Conclusion:

A simple, well-designed tax on cumulative emissions captures **93%** of the potential welfare gains. Adding complex state-dependencies provides only marginal benefits.

The Cost of Political Feasibility

Comparing the physical outcomes of the two approaches reveals a crucial trade-off:

1. Welfare Maximizing
(Unconstrained)

- Aggressive early mitigation.
- Warming stabilizes at $\approx$ **2.7°C**.
- Current generations lose (up to 5%).

2. Pareto Improving
(Constrained)

- Slower initial mitigation path.
- Warming stabilizes at $\approx$ **3.0°C**.
- No generation loses.
- **Crucial: Truncates worst-case damages (insurance).**

Trade-off & Insurance

Ensuring political support requires delaying some mitigation, leading to higher *average* temperatures ($\approx +0.3^{\circ}C$). However, it successfully **eliminates extreme damage scenarios** ($> 15\%$ of GDP), effectively insuring future generations against climate disasters.

One Surrogate, Three Questions

	Session 2: SMM	Appendix: UQ	Today: Design
Model	Brock–Mirman	stochastic IAM with Bayesian learning about the ECS	stochastic OLG IAM with tipping risk
Surrogate inputs	(z, K, β, ϱ)	climate-economy states plus uncertain $\boldsymbol{\theta}$ (9+N states)	OLG IAM state plus 16 policy pseudo-states
QoI	moments m_1 to m_3	$\mathrm{SCC}(\theta_1, \ldots, \theta_N)$	40 cohort welfares
Second surrogate	none needed	GP + Bayesian active learning	GP $\times$40 with BAL
Operator	arg min under CRN	global sensitivity analysis (UE, Sobol', Shapley)	constrained arg max
Output	interior estimate	importance ranking of parameters	Pareto-improving tax
Headline	identification-ridge lesson	ECS and damage-function uncertainty drive the SCC	0.42% aggregate gain

Practical Guidance

- **Train once, then simulate for free.** Surrogate 1 costs about 4h from scratch on a laptop, a few minutes warm-started; once trained, the solution allows us to simulate data essentially for free.
- **When the second surrogate pays off.** Whenever a single QoI point is expensive: expected welfare needs 10,000 Monte Carlo paths per policy ϑ; the GP + BAL layer reduces this to 500 simulations total (450 Latin hypercube + 50 actively learned), at LOO error $\approx 10^{-4}$.
- **A generic blueprint.** A universal framework for **almost any** stochastic heterogeneous-agent model: it allows researchers to both solve global equilibria and derive constrained-optimal policies in settings previously deemed intractable.
- **Production-scale reference.** Chen, Didisheim & Scheidegger (2026, *J. Financial Economics*): exactly this surrogate logic at scale, applied to finance.

Extensions and Open Questions

Extensions on the authors' agenda:

- Analyze richer, non-linear tax functions, solve for the optimal tax, and construct the Pareto frontier.
- Optimal transfers in a "quadratic" (parametrized) tax model.
- Based on high-dimensional model representation (Eftekhari and Scheidegger, 2022), determine the dominant terms that are driving the results.
- Add multiple global regions to the model to study the distributional effects of climate change.

Open questions (bridges between the main part and the appendix; not in the source papers):

- Apply the **quantify** operator to the **design** output: how sensitive is $\vartheta^*(\boldsymbol{\theta})$ to the structural parameters?
- Propagate Session 2-style estimation uncertainty in $\hat{\boldsymbol{\theta}}$ into the SCC.

Conclusion

- **One pipeline, three operators:**
  A scalable ML framework (DEQN + GP + post-processing) supports **estimate** (Session 2: SMM with an interior estimate), **quantify**, and **design** on the same trained surrogate.

- **Quantify (appendix):**
  Uncertainty in the SCC is mainly driven by parametric uncertainty in the ECS and the damage function.

- **Design (main part):**
  Simple linear tax + optimal transfers $\rightarrow$ **Pareto improvement**, with a **0.42%** aggregate welfare gain; added complexity yields only marginal gains (to **0.45%**): simple, implementable rules capture the vast majority of benefits.

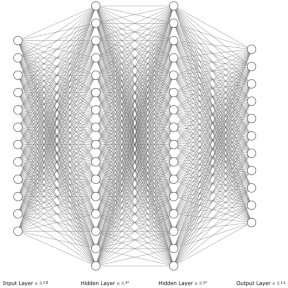
- **Solve once, ask many questions.**

Questions?

Appendix: What is a deep neural net?

$$
\begin{aligned}
\textbf{input} := x &\to {\color{blue}\phi^1(}W^1_{\boldsymbol{\rho}} x + \mathbf{b}^1_{\boldsymbol{\rho}}{\color{blue})} =: \textbf{hidden 1} \\
\to \textbf{hidden 1} &\to {\color{blue}\phi^2(}W^2_{\boldsymbol{\rho}}(\textbf{hidden 1}) + \mathbf{b}^2_{\boldsymbol{\rho}}{\color{blue})} =: \textbf{hidden 2} \\
\to \textbf{hidden 2} &\to {\color{blue}\phi^3(}W^3_{\boldsymbol{\rho}}(\textbf{hidden 2}) + \mathbf{b}^3_{\boldsymbol{\rho}}{\color{blue})} =: \textbf{output}
\end{aligned}
$$

The neural net is then given by the choice of activation functions and the parameters $\boldsymbol{\rho}$.



Appendix: Our loss function

As a loss function, we implement

$$\ell_{\boldsymbol{\rho}} := \frac{1}{N_{\text{path length}}} \sum_{x_i \text{ on sim. path}} \left(\mathbf{G}(x_i, \mathcal{N}_{\boldsymbol{\rho}})\right)^2$$

where we use $\mathcal{N}_{\boldsymbol{\rho}}$ to simulate a path. $\mathbf{G}$ is chosen, such that the true equilibrium policy $\mathbf{f}(x)$ is defined by $\mathbf{G}(x, \mathbf{f}) = 0 \ \forall x$. Therefore, there is no need for labels to evaluate our loss function.

Appendix: Training Deep Equilibrium Nets[3]

1. Simulate a sequence of states $\mathcal{D}^i_{\text{train}} \leftarrow \{x^i_1, x^i_2, \ldots, x^i_T\}$ from the policy encoded by the network parameters $\boldsymbol{\rho}^i$.
2. Evaluate the errors of the equilibrium conditions on the newly generated set $\mathcal{D}_{\text{train}}$.
3. If the error statistics are not low enough:
   3.1 Update the parameters of the neural network with a gradient descent step (or a variant):

$$\rho_k^{i+1} = \rho_k^i - \alpha_{\text{learn}} \frac{\partial \ell_{\mathcal{D}^i_{\text{train}}}(\boldsymbol{\rho}^i)}{\partial \rho_k^i}.$$

   3.2 Set new starting states for simulation: $x_0^{i+1} = x_T^i$.
   3.3 Increase i by one and go back to step 1.

[3]Sample codes here: `https://github.com/sischei/DeepEquilibriumNets`.

Appendix: Architecture of the Deep Neural Network

Hyperparameter	**Value**
Optimizer (LR)	Adam (1e-5)
Batch size	64
NN hidden layers	2
NN units	256
activation function	GeLu
N inputs	$12 + 5 + \vartheta$
N outputs	$2 \cdot (A - 1) = 22$
N epochs per episode	1
N sim episode length	70 (each step is 5y)
N_batches	512

Table: Hyperparameters of neural network.

Appendix: Gaussian Process Regression: Theory

- **Definition:** A GP is a collection of random variables with a joint Gaussian distribution:

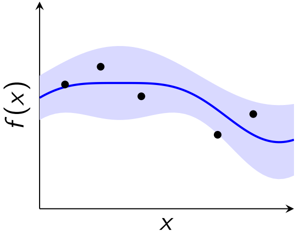
$$f(\mathbf{x}) \sim \mathcal{GP}\Big(m(\mathbf{x}),\ k(\mathbf{x}, \mathbf{x}')\Big).$$

- **Regression:** Given $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$, the posterior for a new $\mathbf{x}_*$ is:

$$f_* \mid \mathbf{x}_*, \mathcal{D} \sim \mathcal{N}(\mu_*, \sigma_*^2).$$

- **Predictive Equations:**

$$\mu_* = \mathbf{k}_*^\top (\mathbf{K} + \sigma_n^2 \mathbf{I})^{-1} \mathbf{y},$$
$$\sigma_*^2 = k_{**} - \mathbf{k}_*^\top (\mathbf{K} + \sigma_n^2 \mathbf{I})^{-1} \mathbf{k}_*.$$



Appendix: Uncertainty (Shock Processes)

- κ_t follows an AR(1) process $\kappa_t = \rho_t \kappa_{t-1} + \epsilon_{\kappa,t}$ with $\kappa_0 = 0.3502$ and $\rho_t = \rho_0 + (\rho_\infty - \rho_0)(1 - \exp(-\delta^\rho 5t))$.
- $\epsilon_{\kappa,t}$ is discretized as:

$$\epsilon_{\kappa,t} = \begin{cases} 0.03 & \text{with prob } 1/3, \\ 0.0 & \text{with prob } 1/3, \\ -0.03 & \text{with prob } 1/3. \end{cases} \tag{13}$$

- TP_t follows a random walk $TP_t = TP_{t-1} + \epsilon_{TP,t}$ with $TP_0 = 3.0$, $\epsilon_{TP,t}$ is discretized as:

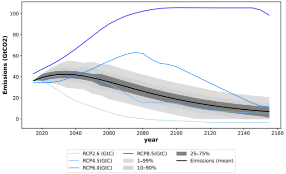
$$\epsilon_{TP,t} = \begin{cases} 0.1 & \text{with prob } 1/3, \\ 0 & \text{with prob } 1/3, \\ -0.1 & \text{with prob } 1/3. \end{cases} \tag{14}$$

Appendix: Calibration

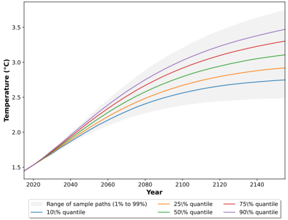
Symbol	Parameter	Value
β	Discount factor	0.9
σ	Relative risk aversion	3
α	Capital share	0.3
δ	Depreciation	0.2
σ_{ccr}	Climate sensitivity	1.7
ψ_1	parameter damage function	13.16
θ_1	Intercept of control cost function	0.7
θ_2	Exponent of control cost function	2.6
κ_0	Initial carbon intensity	0.35032
ρ_0	Initial AR(1) param for carbon intensity	1.08
ρ_∞	Final AR(1) param for carbon intensity	0.91
δ^ρ	AR(1) param depreciation for carbon intensity	0.04
E_0	initial stock of carbon thGtC	0.851
L	Labor (to match initial emissions)	600

Table: Parameterization of the OLG Model.

Appendix: BAU Scenario versus RCP Scenarios



(a) Emissions



(b) Temperature

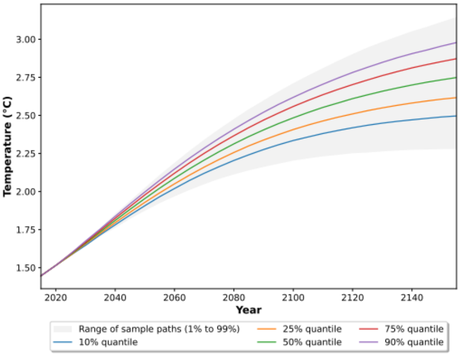
Left: Global CO_2 Emissions

- Comparison of our model's BAU against IPCC Representative Concentration Pathways (RCP).
- Our endogenous emissions path tracks closely with **RCP 4.5**.
- Base year (0) corresponds to 2017.

Right: Global Warming Projections

- Projected mean warming reaches $\approx$ **3°C** over 150 years.
- Substantial uncertainty exists: range of **2.5°C to 3.6°C**.
- Highlights significant tail risk in the absence of policy.

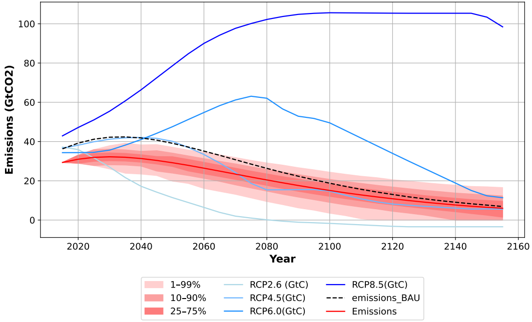
Appendix: Temperature (Scenario 1: Unconstrained)



Outcomes:

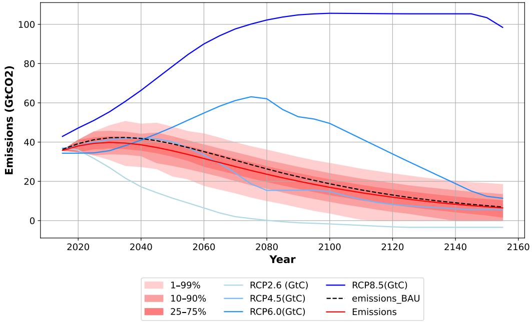
- **Aggressive Mitigation:** Stabilizes mean warming at $\approx$**2.7 °C**.
- **Tail-Risk Insurance:** The 99th percentile is capped at $\approx$3.1 °C.
- **The Mechanism:** Requires immediate, heavy taxation ($\approx$70% average after 150 years).
- **The Cost:** Imposes up to $-$**5% welfare losses** on current older cohorts.

Appendix: Emissions (Welfare Maximizing)



- **Immediate Shock:** Emissions drop $\approx$ **25%** below BAU within the first 50 years.
- **Eliminating the Extremes:** The policy successfully truncates the high-emission tail (the red shaded area).
- **Convergence:** Post-2100, the gap narrows as exogenous carbon intensity declines naturally.

Appendix: Emissions: Mitigating Risk (Scenario 2)



- **Modest Average Reduction:** The mean emissions path is not drastically lower than BAU (required to preserve current consumption).
- **Tail Truncation:** The policy's primary physical impact is eliminating the high-emissions upside.
- **Result:** The 99^{th} percentile of emissions drops below 40 $GtCO_2$ by 2070 (20 years earlier than BAU).

Appendix: Sampling Domain

Model	Planner Target	Parameter	$\underline{\vartheta}_i$	$\bar{\vartheta}_i$
Linear in E	Max. Welfare	ϑ_0	-2.0	2.0
	Max. Welfare	ϑ_E	-2.0	2.0
	Max. Welfare	τ	0.0	1.5
Linear in E + Transfers	Pareto	ϑ_0	-0.5	0.1
	Pareto	ϑ_E	0.0	0.5
	Pareto	τ	0.0	0.8
Full linear + Transfers	Pareto	ϑ_0	-1.0	0.5
	Pareto	ϑ_E	-0.5	1.0
	Pareto	ϑ_κ	-0.6	0.6
	Pareto	ϑ_{TP}	-1.0	1.0
	Pareto	τ	0.0	0.8

Table: Parameter bounds across models and objectives.

Appendix: An Incomplete List of Related Literature

Climate Risk in OLG Models

Existing work studies intergenerational conflict and climate damages (Kotlikoff et al., 2021a; Weitzman, 2012; Karp and Rezai, 2014).

+ **Our Gap:** Most rely on linearization or ignore the *political constraint* of Pareto improvements under uncertainty (e.g., tipping points).

Optimal Ramsey Policy with Heterogeneity

Established frameworks for fiscal/monetary policy in HA models (Bhandari et al., 2013; Nuño and Thomas, 2020; Douenne et al., 2024).

+ **Our Gap:** We enforce *simple rules* (linear taxes) to make the policy implementable and interpretable.

Deep Learning for High-Dimensional Dynamic Stochastic Models

Using Neural Networks (DEQN) and Surrogates to beat the curse of dimensionality (Azinovic et al., 2022; Friedl et al., 2023).

+ **Our Innovation:** A 3-step framework: DEQN (Global Solution) $\rightarrow$ GP Surrogate $\rightarrow$ Constrained Optimization.

Appendix: How Uncertain Is the Social Cost of Carbon?

The *quantify* operator on the same two-surrogate pipeline

Integrated Assessment Models (IAMs)

- Pioneered by Nordhaus (DICE, RICE): Quantitative, often numerical.
- Key components:
  - (Simplified) climate system.
  - (Stylized) economic model of emissions AND damages.
- Economic model: needs to be dynamic, forward-looking, possibly allowing stochasticity (temperature variations, disasters, TFP), and potentially heterogeneous agents ("who is how impacted?").

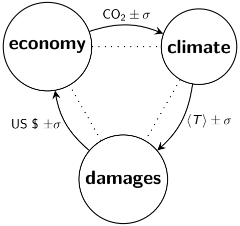


Figure: Stylized representation of an IAM.

A Stochastic IAM with Bayesian Learning about the ECS

$$V(\boldsymbol{X}_t)^{1-1/\psi} = \max_{C_t, K_{t+1}, \mu_t} \left\{ \left(\frac{C_t}{L_t}\right)^{1-1/\psi} L_t + e^{-\rho} \mathbb{E}_t \left[V(\boldsymbol{X}_{t+1})^{1-\gamma} \right]^{\frac{1-1/\psi}{1-\gamma}} \right\}$$

$$\begin{aligned}
\text{s.t.} \quad & (1-\Theta(\mu_t))\,\Omega_t(T_{\mathrm{AT},t})\,K_t^{\alpha}(A_t L_t)^{1-\alpha} - C_t - I_t = 0 \\
& (1-\delta)K_t + I_t - K_{t+1} = 0 \\
& 1-\mu_t \geq 0 \\
& (1-\phi_{12})M_{\mathrm{AT},t} + \phi_{21}M_{\mathrm{UO},t} + (1-\mu_t)\sigma_t K_t^{\alpha}(A_t L_t)^{1-\alpha} + E_{\mathrm{Land},t} - M_{\mathrm{AT},t+1} = 0 \\
& \phi_{12}M_{\mathrm{AT},t} + (1-\phi_{21}-\phi_{23})M_{\mathrm{UO},t} + \phi_{32}M_{\mathrm{LO},t} - M_{\mathrm{UO},t+1} = 0 \\
& \phi_{23}M_{\mathrm{UO},t} + (1-\phi_{32})M_{\mathrm{LO},t} - M_{\mathrm{LO},t+1} = 0 \\
& (1-\varphi_{21}-\varphi_{1C})T_{\mathrm{AT},t} + \varphi_{21}T_{\mathrm{OC},t} + \varphi_1\left(F_{\mathrm{2xco2}}\log_2\left(\frac{M_{\mathrm{AT},t}}{M_{\mathrm{AT}}^*}\right) + F_{\mathrm{EX},t}\right) + \\
& \qquad \varphi_{1C}\tilde{f}_{t+1}T_{\mathrm{AT},t} + \tilde{\epsilon}_{T,t+1} - T_{\mathrm{AT},t+1} = 0 \\
& \varphi_4 T_{\mathrm{AT},t} + (1-\varphi_4)T_{\mathrm{OC},t} - T_{\mathrm{OC},t+1} = 0 \\
& \frac{S_{\epsilon_T}\mu_{f,t} + \varphi_{1C}T_{\mathrm{AT},t}\left(\varphi_{1C}T_{\mathrm{AT},t}\tilde{f}_{t+1} + \tilde{\epsilon}_{T,t+1}\right)S_{f,t}}{S_{\epsilon_T} + (\varphi_{1C}T_{\mathrm{AT},t})^2 S_{f,t}} - \mu_{f,t+1} = 0 \\
& \frac{S_{\epsilon^T}S_{f,t}}{S_{\epsilon^T} + (\varphi_{1C}T_{\mathrm{AT},t})^2 S_{f,t}} - S_{f,t+1} = 0 \\
& \tilde{f}_{t+1} \sim \mathcal{N}\left(\mu_{f,t}, S_{f,t}, \underline{f}, \overline{f}\right), \tilde{\epsilon}_{T,t+1} \sim \mathcal{N}(0, S_{\epsilon^T})
\end{aligned}$$

- $X_t = \left[k_t, M_{\mathrm{AT},t}, M_{\mathrm{UO},t}, M_{\mathrm{LO},t}, T_{\mathrm{AT},t}, T_{\mathrm{OC}}, \mu_{f,t}, S_{f,t}, t; \theta\right]$.
- Minimal model has a **9+N**-dimensional state space.
- Uncertain equilibrium climate sensitivity following Roe and Baker (2007): climate feedback $f \sim \mathcal{N}(\mu_f, S_f)$; details in the appendix.

Parameter Ranges for the UQ Analysis

Parameter	θ_i^0	$\underline{\theta}_i$	$\overline{\theta}_i$	Source etc.
ρ	0.015	0.01	0.02	Stern (2007)
γ	10.	5.0	10.0	Jensen and Traeger (2014) and Cai and Lontzek (2019)
ψ	1.5	0.5	2.0	–//–
π_2	0.00236	0.002	0.008	Nordhaus (2017a) and Weitzman (2012)
$\mu_{f,0}$	0.65	0.45	0.73	Roe and Baker (2007)
$S_{f,0}$	0.0169	0.01	0.04	Roe and Baker (2007)

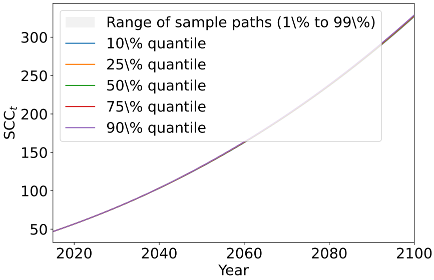
These are the uncertain $\boldsymbol{\theta}$ entering the surrogate as pseudo-states.

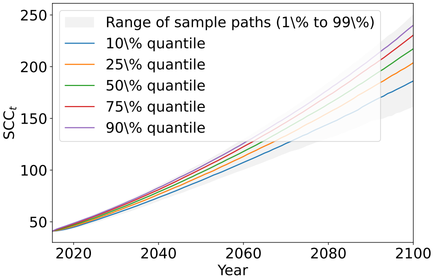
Instantiating the Pipeline: Parameters as Pseudo-States

- DEQNs can solve nonlinear stochastic models globally, even if geometry is oddly-shaped, in minutes to hours on a laptop; state spaces up to dozens of variables.
- Add pseudo-states for uncertainty quantification at a small extra cost: only one **single** model evaluation needed: here the pseudo-states are the structural parameters $\boldsymbol{\theta}$, the move Session 2 made with ϱ and β.
- The computed policies are functions of this extended state space; to obtain QoIs as a function of the parameters, simulate the economy with the derived policy functions to compute the SCC at N points.
- The remaining UQ tasks are just "post-processing", and come at low computational costs, as we can query a **surrogate** (Surrogate 2, fitted to those N points).

The QoI: Social Cost of Carbon [$/tonC]

The SCC: the marginal loss caused by an extra ton of CO_2 emissions; the level of the carbon tax is based on it. Computed here from surrogate simulations.





Model with no learning (left) and learning (right); SCC behaves like a random variable, as Temperature and ECS are stochastic (our results confirm Cai and Lontzek (2019)).

Global Sensitivity Analysis: 3 Quantities

- **How are quantitative results of IAMs sensitive to a specific parameterization**? A **importance ranking** informs the researcher on which parts to focus when calibrating or extending a model, or the policymaker on which parameters need further scrutiny.
- "One-at-a-time" approaches tend to be unstructured and are only **local**: highly dependent on the chosen parameter values. We aim at determining which input parameters (or combinations) contribute the most to the uncertainty of the QoI, globally.
- Measures we use:
  - **Sobol' index**: prioritization of the uncertain parameters based on the outcome variance explained.
  - **Shapley value**: identifies how much model variance can be attributed to the uncertainty in input parameters.
  - **Univariate effect**: shows how each parameter affects the outcomes.

$\rightarrow$ All three are computed on a cheap-to-evaluate **surrogate model** based on Gaussian Processes (Scheidegger and Bilionis, 2019), e.g., SCC$(\theta_1, \theta_2, ..., \theta_N)$.

Univariate Effect (UE)

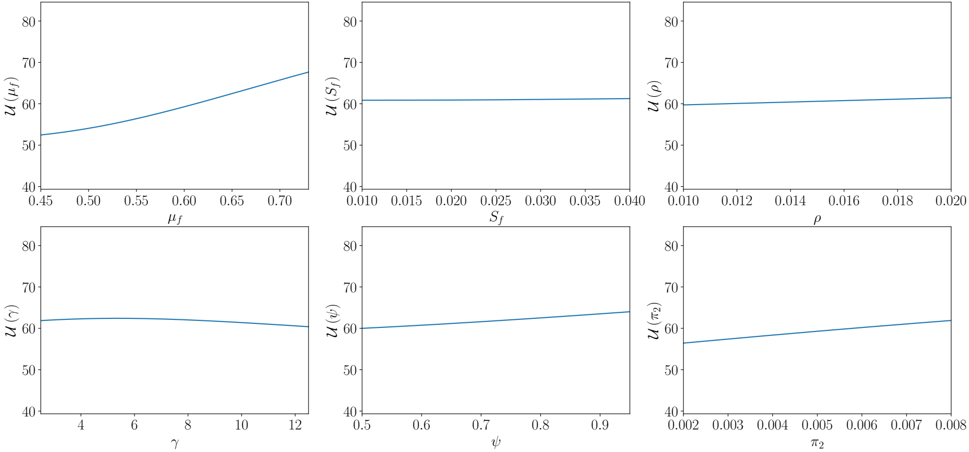
- UE: measures the non-linear relationship between the target QoI and an uncertain input parameter.
- UE is the conditional expectation, integrating over the other uncertain parameters θ_{-i}, of the QoI as a function of an input parameter θ_i fixed at an arbitrary value t:[4]

$$\mathcal{M}_i^1(t) = \mathbb{E}_{\theta_{-i}}\left[Y \mid \theta_i = t\right].$$

→ Sobol' indices do not include information about the direction in which a parameter affects the quantities of interest.
→ Univariate effects help to find regions of high and low sensitivity, and can be interpreted as a robust direction of change under parameter uncertainty.

[4] In the source figures the fixed value is denoted ϑ_i; we reserve ϑ for policy parameters throughout this lecture.

Univariate Effects on the SCC in 2020 (Learning Model)



Sobol' Indices

- Consider a generic model $\mathcal{M}(\cdot)$ with N uncertain input parameters θ and output y summarizing the QoI (e.g., SCC):

$$\theta \in \mathcal{D}_\theta \subset \mathbb{R}^N \mapsto y = \mathcal{M}(\theta_1, \theta_2, \cdots, \theta_N) \in \mathbb{R}^Q.$$

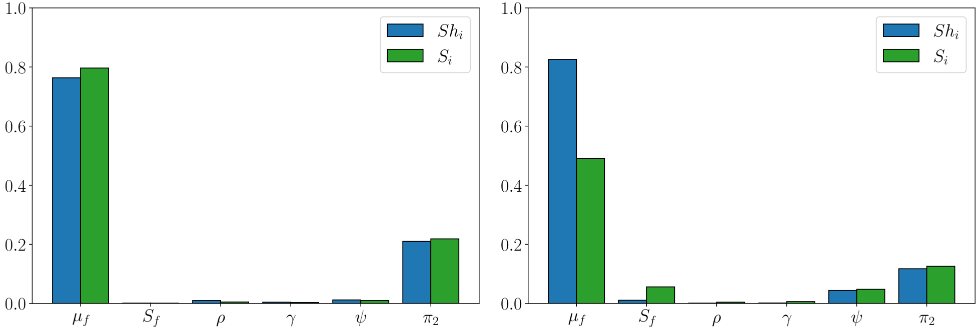
- The first-order Sobol' indices are defined as:[5]

$$S_i \equiv \frac{\mathrm{Var}_{\theta_i}\left[\mathbb{E}_{\Theta\setminus\theta_i}[y \mid \theta_i]\right]}{\mathrm{Var}_\Theta[y]}$$

- The numerator measures the first-order effect of θ_i on the model output y; normalizing by the total model variance $\mathrm{Var}_\Theta[y]$ scales the index to $[0, 1]$.
- **Screening interpretation:** the first-order Sobol' index indicates by what percentage the total variance would be reduced, should the parameter θ_i be perfectly known and set to a fixed value. If it is small (in practice $< 1\%$), θ_i can be set to a deterministic value without changing the distribution of the QoI.

[5] For simplicity, we assume $Q = 1$.

Importance of Parameters for SCC in 2020



UQ results (Shapley values and first-order Sobol' indices) for the model without learning (left) and with learning (right) in 2020.

UQ Takeaways: What Drives SCC Uncertainty

- Uncertainty in SCC is mainly driven by parametric uncertainty in the ECS and the damage function.
- Overall, highly nuanced results.
- The quantify operator ran entirely in post-processing: the equilibrium was solved exactly once.
- **Some open-source codes here**:
  - `https://github.com/ClimateChangeEcon/Climate_in_Climate_Economics`
  - `https://github.com/sischei/DeepEquilibriumNets`

Appendix: The blue marble in 5 state variables

- Climate module: DICE-2016 functional form (Nordhaus, 2017b), calibrated to CMIP5 output (Folini et al., 2021).
- **5 state variables** = 3 carbon reservoirs + 2 temperature layers.

Carbon cycle: 3 reservoirs: Atmosphere (A), Upper ocean (U), Lower ocean (L).

$$
\begin{aligned}
M_t &= (M_{A,t}, M_{U,t}, M_{L,t})^\top \\
M_{t+1} &= (I + \Delta_t B)\, M_t + \Delta_t E_t \\
E_t &= E_{\mathrm{ind},t} + E_{\mathrm{Land}}
\end{aligned}
$$

Energy balance: 2 layers: atmosphere T^{AT}, ocean T^{OC}.

$$
\begin{aligned}
T^{\mathrm{AT}}_{t+1} &= T^{\mathrm{AT}}_t + \Delta_t c_1 \big(F_t - \lambda T^{\mathrm{AT}}_t - c_3 (T^{\mathrm{AT}}_t - T^{\mathrm{OC}}_t)\big) \\
T^{\mathrm{OC}}_{t+1} &= T^{\mathrm{OC}}_t + \Delta_t c_4 (T^{\mathrm{AT}}_t - T^{\mathrm{OC}}_t) \\
F_t &= F_{\mathrm{2XCO2}} \frac{\log(M^{\mathrm{AT}}_t / M^{\mathrm{AT}}_{\mathrm{EQ}})}{\log 2} + F^{\mathrm{EX}}_t
\end{aligned}
$$

$\rightarrow$ Emissions E_t (economic activity + exogenous land source) raise atmospheric carbon M_{AT}, which drives the forcing F_t and hence temperature T^{AT}.

Appendix: A Source of Uncertainty: Equilibrium Climate Sensitivity

IPCC AR5 gives a 'likely range' of [1.5K, 4.5K]

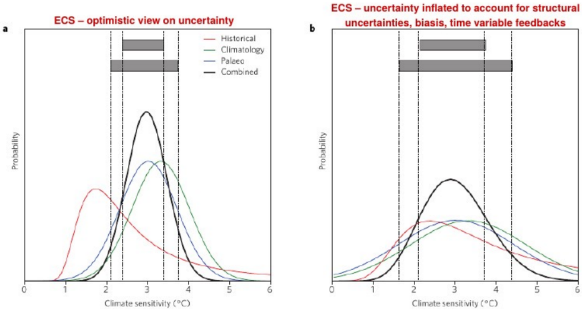
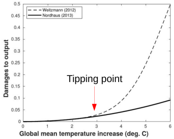


Figure 5 | Illustrative example of combining multiple constraints for climate sensitivity. Overall PDFs are the products of three PDFs based on the historical warming, climatological constraints on mostly GCMs, and palaeoclimate. Grey ranges at the top indicate a 'likely' (66%) and 'very likely' (90%) combined range. **a**,**b**, Constraint based on an optimistic interpretation of uncertainty ranges and assuming full independence (**a**) and on inflated ranges (**b**) to account for structural uncertainties, and with historical estimates inflated and scaled up to account for observation biases, and feedbacks varying from historical to future and across forcings. See Methods for details.

Knutti et al. 2017, Nature, doi: 10.1038/NGEO3017

Appendix: Another Source of Uncertainty: Damage Functions

See also Tol (2009), Hänsel et al. (2020)



- Weitzman (2012):

$$\Omega_t\left(T_{\mathrm{AT},t}\right)=\frac{1}{1+\left(\frac{1}{\psi_1}T_{\mathrm{AT},t}\right)^2+\left(\frac{1}{2\cdot TP_t}T_{\mathrm{AT},t}\right)^{6.754}}.$$

- Nordhaus (2013):

$$\Omega_t^N\left(T_{\mathrm{AT},t}\right)=\frac{1}{1+\pi_1 T_{\mathrm{AT},t}+\pi_2 T_{\mathrm{AT},t}^2}.$$

- Degree of convexity is of key importance in determining the optimal taxes, not only the level.
- Tipping points: include losing much of the Amazon rain forest, faster onset of El Nino, the reversal of the Gulf Stream, etc.

Appendix: Quantifying uncertainty about ECS

- Temperature equation:

$$T_{\mathrm{AT},t+1} = \left(1 - \varphi_{21} - \varphi_1 \frac{F_{\mathrm{2xco2}}}{\Delta T_{\mathrm{AT},\times 2}}\right) T_{\mathrm{AT},t} + \varphi_{21} T_{\mathrm{OC},t} + \tag{15}$$

$$\varphi_1 \left(F_{\mathrm{2xco2}} \log_2 \left(\frac{M_{\mathrm{AT},t}}{M^*_{\mathrm{AT}}}\right) + F_{\mathrm{EX},t}\right) \tag{16}$$

- Follow Roe and Baker (2007) to make ECS stochastic.
- $\mathrm{ECS} = \frac{\lambda_0}{1-f} F_{\mathrm{2XCO2}}$, with climate feedback parameter $f \sim \mathcal{N}(\mu_f, S_f)$.
- This is controversial, see Zaliapin and Ghil (2010) and Roe and Baker (2011) and Zaliapin and Ghil (2011).

Appendix: Bayesian learning mechanics

- Temperature evolves as:

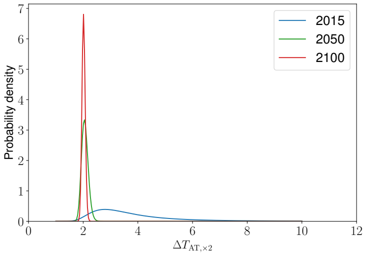
$$T_{\mathrm{AT},t+1} = \left(\varphi_1 \frac{F_{\mathrm{2xco2}}}{\Delta T^0_{\mathrm{AT},\times 2}} \tilde{f}_{t+1} + 1 - \varphi_{21} - \varphi_1 \frac{F_{\mathrm{2xco2}}}{\Delta T^0_{\mathrm{AT},\times 2}}\right) T_{\mathrm{AT},t}$$
$$+ \varphi_{21} T_{\mathrm{OC},t} + \varphi_1 \left(F_{\mathrm{2xco2}} \log_2 \left(\frac{M_{\mathrm{AT},t}}{M^*_{\mathrm{AT}}}\right) + F_{\mathrm{EX},t}\right) + \tilde{\epsilon}_{T,t+1}$$

  where $\tilde{f}_{t+1} \sim \mathcal{N}\left(\mu_{f,t}, S_{f,t}, \underline{f}, \overline{f}\right)$: uncertain climate sensitivity, $\tilde{\epsilon}_{T,t+1}$: shock to the temperature.

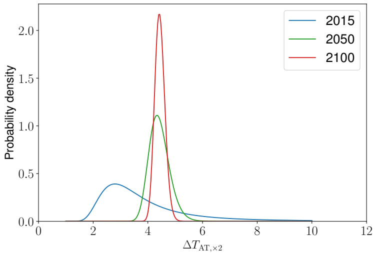
- The agent observes a realisation of the shocks in climate sensitivity and temperature at the beginning of the period t before the decisions are made.
- Under Gaussian assumption analytic formulas for updating:

$$\mu_{f,t+1} = \frac{S_{\epsilon_T} \mu_{f,t} + \varphi_{1C} T_{\mathrm{AT},t} \left(\varphi_{1C} T_{\mathrm{AT},t} \tilde{f}_{t+1} + \tilde{\epsilon}_{T,t+1}\right) S_{f,t}}{S_{\epsilon_T} + \left(\varphi_{1C} T_{\mathrm{AT},t}\right)^2 S_{f,t}}$$
$$S_{f,t+1} = \frac{S_{\epsilon_T} S_{f,t}}{S_{\epsilon_T} + \left(\varphi_{1C} T_{\mathrm{AT},t}\right)^2 S_{f,t}}$$

Appendix: Tail learning



(a) $\Delta T^*_{\mathrm{AT},\times 2} = 2.0$



(b) $\Delta T^*_{\mathrm{AT},\times 2} = 4.5$

Figure: The posterior probability density function of $\Delta T_{\mathrm{AT},\times 2}$ is shown when the true equilibrium sensitivity $\Delta T^*_{\mathrm{AT},\times 2}$ is set to 2.0 and 4.5, respectively.

- Social planner learns the upper tail of the prior distribution very quickly (less than a decade for 99% percentile; ECS < 3.42).
- In the case where the ECS is set to 4.5, that is, relatively high, learning slows as the Bayes rule requires more observations to move the mean estimate from the prior ECS to the true high value.

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